NB39: MASTER - Reversed-Distribution Egitim Deneyi

SEED=42 | TEST_SIZE=0.2 | 5 senaryo x 4 model

0. Motivasyon ve Hipotez

NB38 en iyi sonuc: V1_calib F1_8020=0.6165 (heterojen stacking + isotonic kalibrasyon).
Calibrate-then-shift MASTER da ek kazanc vermedi. Precision=0.55 darbogazda.

Hipotez: Modeli final test dagilimina (%80 benign / %20 pathogenic) yakin bir egitim setiyle
egitmek, post-hoc duzeltmelere gore daha etkili olabilir.
Kanit: KANSER panelinde P9_REVERSE_6040 F1_8020 yi 0.69->0.73 e tasidi.

1. Deney Tasarimi

Senaryo	Kompozisyon	n_train	Benign Frac
S0_baseline	Orijinal (~73:27 patho:ben)	2344	0.267
S1_6040	%60 benign / %40 patho	1041	0.600
S2_8020	%80 benign / %20 patho	780	0.800
S3_7030	%70 benign / %30 patho	892	0.701
S4_weighted	Tum veri + class_weight	2344	agirlikli

Veri: MASTER (n=2931, pos=2149, neg=782)
Split: %80/20 stratified (SEED=42)
Test: 587 (NB38 ile birebir ayni)
Feature temizlik: sabit=57, ozdes=583, kalan=287
Missing: M3 (is_missing >%50 + medyan) - her senaryo icin AYRI fit
Modeller: LightGBM, BalancedBag, RF, CatBoost (NN haric)

2. S0_baseline Sonuclari

Model	F1_8020	CI_95	MCC	Prec	Rec	F1_50	Gap
lgbm	0.5511	[0.453-0.640]	0.426	0.462	0.687	0.779	+0.100
balbag	0.5747	[0.485-0.646]	0.459	0.484	0.712	0.796	+0.097
rf	0.6087	[0.497-0.696]	0.504	0.545	0.693	0.793	+0.094
catboost	0.5810	[0.491-0.663]	0.467	0.491	0.715	0.799	+0.075

2. S1_6040 Sonuclari

Model	F1_8020	CI_95	MCC	Prec	Rec	F1_50	Gap
lgbm	0.5804	[0.491-0.656]	0.466	0.513	0.673	0.776	+0.220
balbag	0.6379	[0.519-0.716]	0.542	0.583	0.708	0.802	+0.253
rf	0.6065	[0.489-0.680]	0.501	0.545	0.688	0.788	+0.259
catboost	0.5963	[0.480-0.665]	0.489	0.544	0.664	0.774	+0.198

2. S2_8020 Sonuclari

Model	F1_8020	CI_95	MCC	Prec	Rec	F1_50	Gap
lgbm	0.5146	[0.448-0.600]	0.380	0.389	0.763	0.809	+0.431
balbag	0.5486	[0.424-0.643]	0.429	0.508	0.602	0.716	+0.286
rf	0.4876	[0.371-0.587]	0.338	0.384	0.671	0.750	+0.482
catboost	0.4915	[0.424-0.552]	0.352	0.356	0.797	0.826	+0.253

2. S3_7030 Sonuclari

Model	F1_8020	CI_95	MCC	Prec	Rec	F1_50	Gap

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lgbm	0.5700	[0.446-0.657]	0.454	0.515	0.643	0.757	+0.284
balbag	0.5738	[0.439-0.652]	0.461	0.531	0.629	0.750	+0.253
rf	0.5621	[0.429-0.656]	0.442	0.487	0.669	0.769	+0.328
catboost	0.5823	[0.465-0.671]	0.472	0.542	0.634	0.750	+0.226

2. S4_weighted Sonuclari

Model	F1_8020	CI_95	MCC	Prec	Rec	F1_50	Gap
lgbm	0.5917	[0.483-0.673]	0.485	0.552	0.642	0.751	+0.147
balbag	0.5747	[0.485-0.646]	0.459	0.484	0.712	0.796	+0.097
rf	0.4708	[0.383-0.560]	0.318	0.393	0.592	0.706	+0.107
catboost	0.5911	[0.508-0.665]	0.481	0.499	0.729	0.810	+0.102

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SEED=42 | TEST_SIZE=0.2 | 5 senaryo x 4 model

3. Buyuk Karsilastirma Tablosu (tum senaryolar)

Model	n_train	F1_8020	CI_95	MCC	Prec	Rec	Gap
S1_6040/balbag	1041	0.6379	[0.519-0.716]	0.542	0.583	0.708	+0.253
NB38_V1_calib	2345	0.6165		0.000	0.000	0.000	+0.000
S0_baseline/rf	2344	0.6087	[0.497-0.696]	0.504	0.545	0.693	+0.094
S1_6040/rf	1041	0.6065	[0.489-0.680]	0.501	0.545	0.688	+0.259
S1_6040/catboost	1041	0.5963	[0.480-0.665]	0.489	0.544	0.664	+0.198
S4_weighted/lgbm	2344	0.5917	[0.483-0.673]	0.485	0.552	0.642	+0.147
S4_weighted/catboost	2344	0.5911	[0.508-0.665]	0.481	0.499	0.729	+0.102
S3_7030/catboost	892	0.5823	[0.465-0.671]	0.472	0.542	0.634	+0.226
S0_baseline/catboost	2344	0.5810	[0.491-0.663]	0.467	0.491	0.715	+0.075
S1_6040/lgbm	1041	0.5804	[0.491-0.656]	0.466	0.513	0.673	+0.220
S4_weighted/balbag	2344	0.5747	[0.485-0.646]	0.459	0.484	0.712	+0.097
S0_baseline/balbag	2344	0.5747	[0.485-0.646]	0.459	0.484	0.712	+0.097
S3_7030/balbag	892	0.5738	[0.439-0.652]	0.461	0.531	0.629	+0.253
S3_7030/lgbm	892	0.5700	[0.446-0.657]	0.454	0.515	0.643	+0.284
S3_7030/rf	892	0.5621	[0.429-0.656]	0.442	0.487	0.669	+0.328
S0_baseline/lgbm	2344	0.5511	[0.453-0.640]	0.426	0.462	0.687	+0.100
S2_8020/balbag	780	0.5486	[0.424-0.643]	0.429	0.508	0.602	+0.286
S2_8020/lgbm	780	0.5146	[0.448-0.600]	0.380	0.389	0.763	+0.431
S2_8020/catboost	780	0.4915	[0.424-0.552]	0.352	0.356	0.797	+0.253
S2_8020/rf	780	0.4876	[0.371-0.587]	0.338	0.384	0.671	+0.482
S4_weighted/rf	2344	0.4708	[0.383-0.560]	0.318	0.393	0.592	+0.107

4. Hipotez: Threshold Yakinsama

Hipotez: S1/S2/S3 te thr_raw ve thr_8020 arasindaki fark kuculmeli,
cunku egitim dagilimi zaten %80/20 ye yakin.

Threshold farklari:

Senaryo/Model	thr_raw	thr_8020	|Fark|
S0_baseline/lgbm	0.360	0.750	0.390
S0_baseline/balbag	0.220	0.700	0.480
S0_baseline/rf	0.380	0.710	0.330
S0_baseline/catboost	0.360	0.630	0.270
S1_6040/lgbm	0.470	0.545	0.075
S1_6040/balbag	0.380	0.550	0.170
S1_6040/rf	0.370	0.480	0.110
S1_6040/catboost	0.480	0.550	0.070
S2_8020/lgbm	0.240	0.250	0.010
S2_8020/balbag	0.520	0.520	0.000
S2_8020/rf	0.350	0.325	0.025
S2_8020/catboost	0.380	0.425	0.045
S3_7030/lgbm	0.370	0.490	0.120
S3_7030/balbag	0.530	0.585	0.055
S3_7030/rf	0.390	0.420	0.030
S3_7030/catboost	0.520	0.520	0.000
S4_weighted/lgbm	0.170	0.700	0.530
S4_weighted/balbag	0.220	0.700	0.480

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S4_weighted/rf	0.280	0.610	0.330
S4_weighted/catboost	0.310	0.720	0.410

5. Precision Analizi

NB38 de precision=0.55 darbogazdi. Reversed-distribution ile precision artisi beklenir (model benign i daha iyi ogrenir, FP azalir).

Senaryo bazinda ortalama precision:

S0: avg precision_8020 = 0.4957

S1: avg precision_8020 = 0.5463

S2: avg precision_8020 = 0.4092

S3: avg precision_8020 = 0.5187

S4: avg precision_8020 = 0.4820

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6. Sonuc ve Karar

En iyi model: S1_6040/balbag (F1_8020=0.6379)
NB38 referans: V1_calib = 0.6165
Iyilestirme: +0.0214

SONUC: Reversed-distribution egitim MASTER icin CALISTI.
En iyi senaryo/model: S1_6040/balbag. Bu yaklasim final model adayi.

Reversed (S1_6040/balbag: 0.6379) S4_WEIGHTED (0.5917) yi gecti.
Gercek resample, agirliklamadan daha etkili - karar sinirini dogrudan kaydiriyor.

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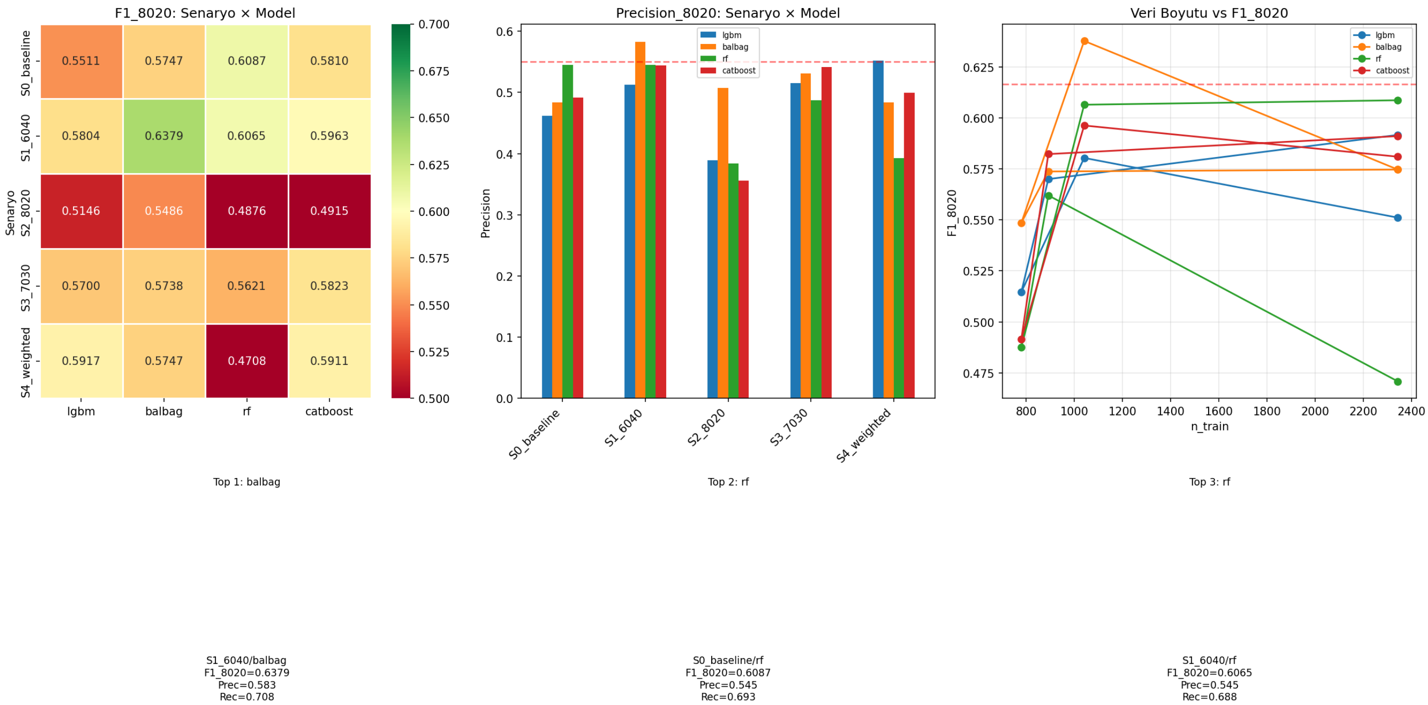
Gorsel: all_experiments.png



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Gorsel: threshold_convergence.png